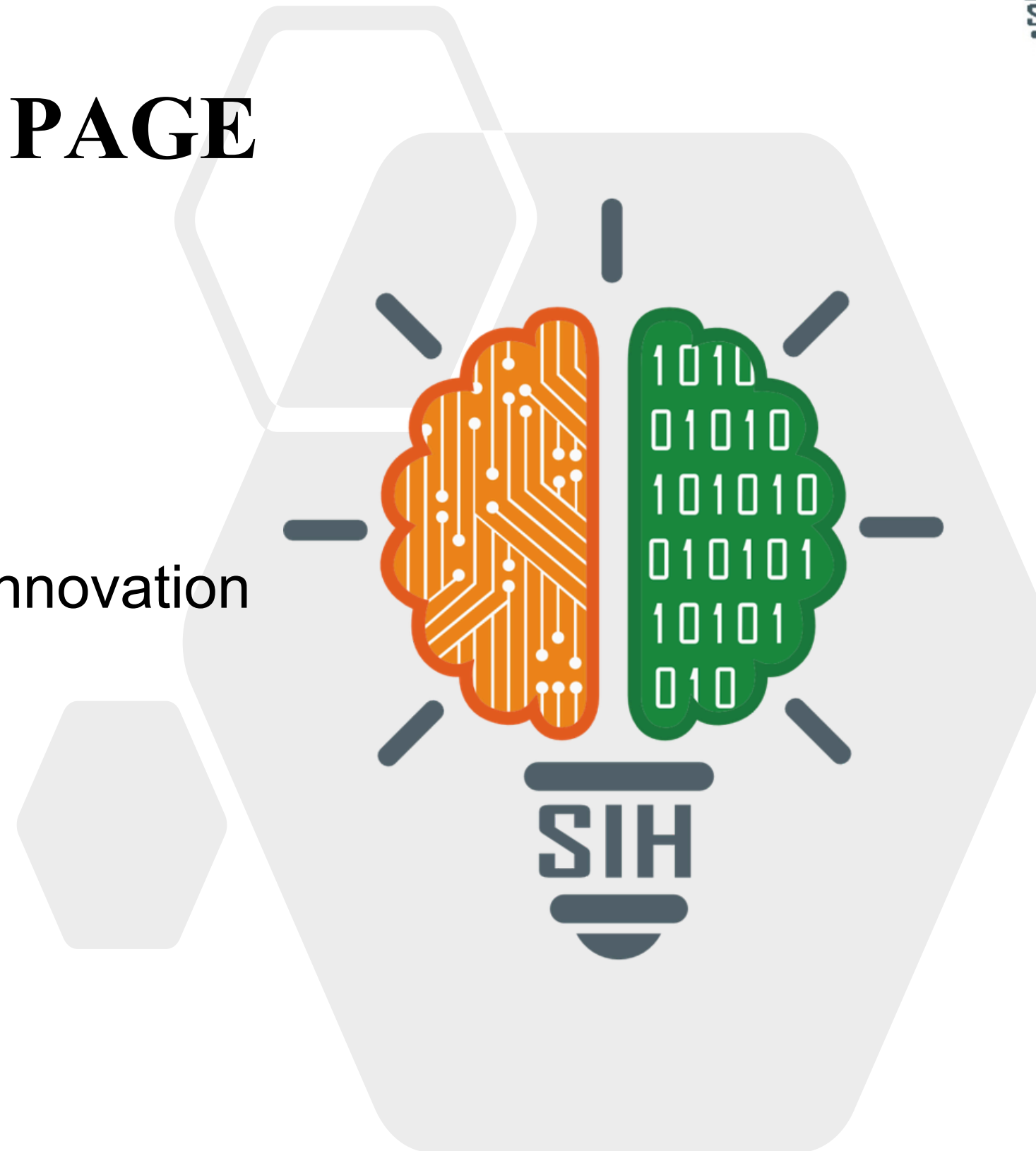


TITLE PAGE

- **Problem Statement ID –** SIH26204
- **Problem Statement Title-** Student Innovation
- **Theme-** Miscellaneous
- **PS Category-** Software
- **Team ID-**
- **Team Name (Registered on portal)-** STACK STORM



PAVITRA CYCLE : AI- POWERED RESPECTFUL DISPOSAL & RECOVERY

SYSTEM

● Problem Explanation

- ▲ Religious offerings often contain materials with **no clear disposal pathway**.
- ▲ Existing collection and recovery efforts are fragmented and **difficult for users to access**.

● Proposed Solution

- ▲ Scan → Identify → Guide → Connect → Track.
- ▲ Ai Identifies the material and recommends the appropriate disposal/ recovery pathway.

● Innovation and Uniqueness

- ▲ AI-based religious material identification.
- ▲ Connects Devotees → Temple → Collectors → Recovery Partners
- ▲ An **intelligent coordination layer** connecting existing collection and recovery systems.

✗ The Problem

Where does our sacred offerings go?



Temples



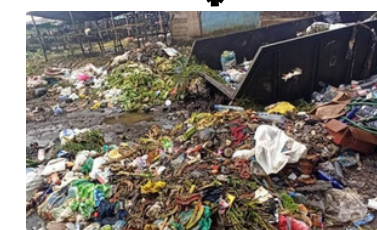
Religious offerings



Mixed with regular waste



Water pollution



Improper disposal



Loss of valuable resources

PAVITRA CYCLE

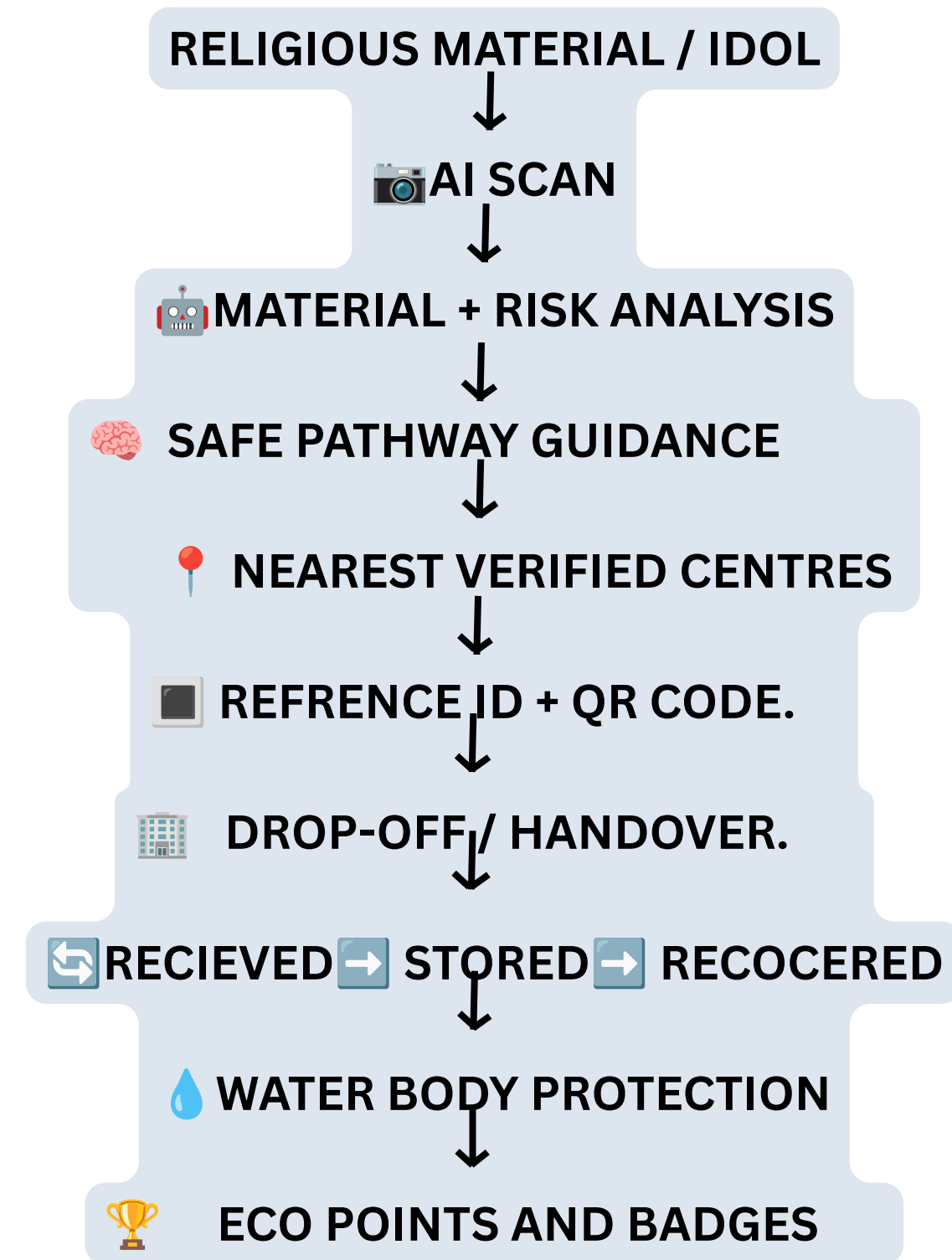


Technologies Used

- ▲  **Computer Vision** - Religious material identification
- ▲  **Google Maps API** - Verified drop-off centres
- ▲  **QR + Unique ID** - Digital handover pass
- ▲  **Firebase/ Database** - Recovery tracking
- ▲  **Recommendation Engine** - Safe disposal guidance
- ▲  **Gaimification System** - Ecosystem & Badges

Methadology

 Scan →
  Analyse. →
  Guide →
  Locate →
  QR Handover →
  Recover & Track →
  Reward



✓ Feasibility

- ▲ Uses **available AI , Maps, QR & Firebase** technologies.
- ▲ Can start as a **city level pilot** and scale gradually.

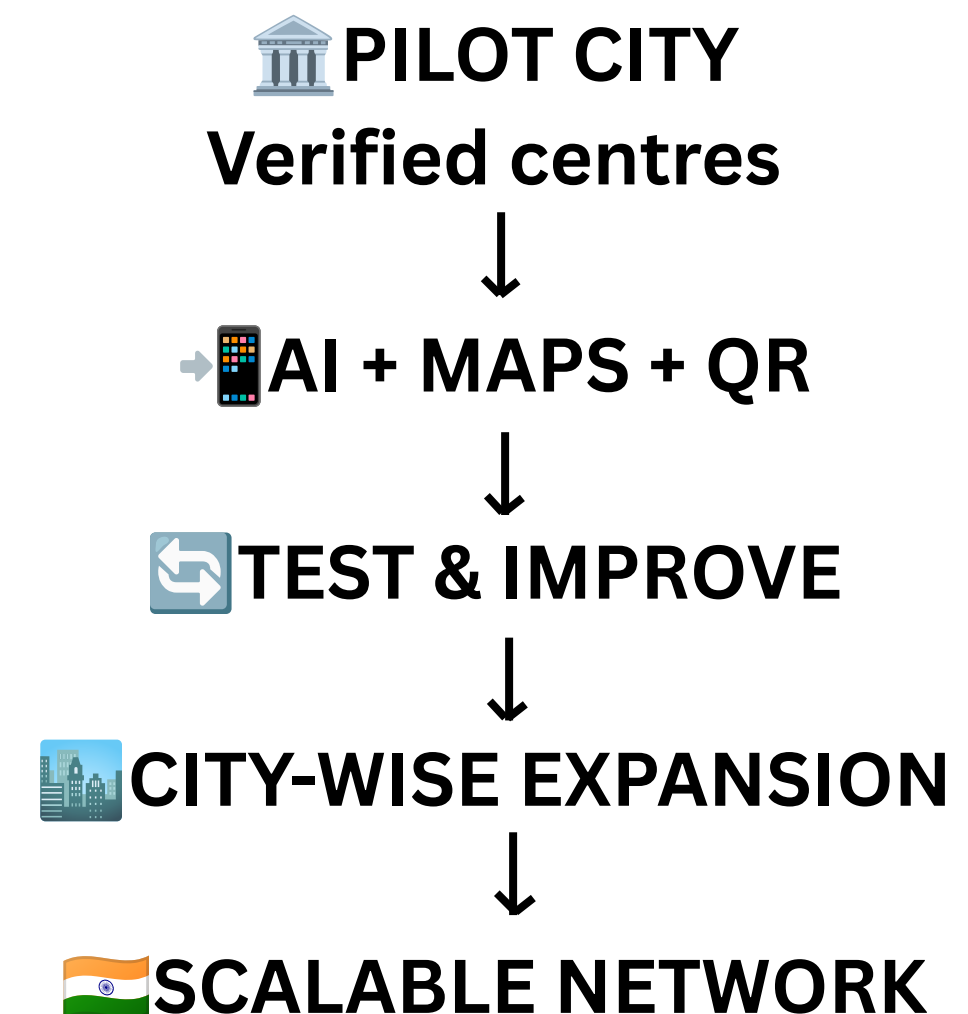
⚠ Challenges

- ▲ Limited verified centres/ partners.
- ▲ AI accuracy for mixed materials.

🔧 Solutions

- ▲ Partner verification+ regular updates.
- ▲ User conformation for low confidence AI results.

PILOT TODAY, PRESERVE TOMORROW



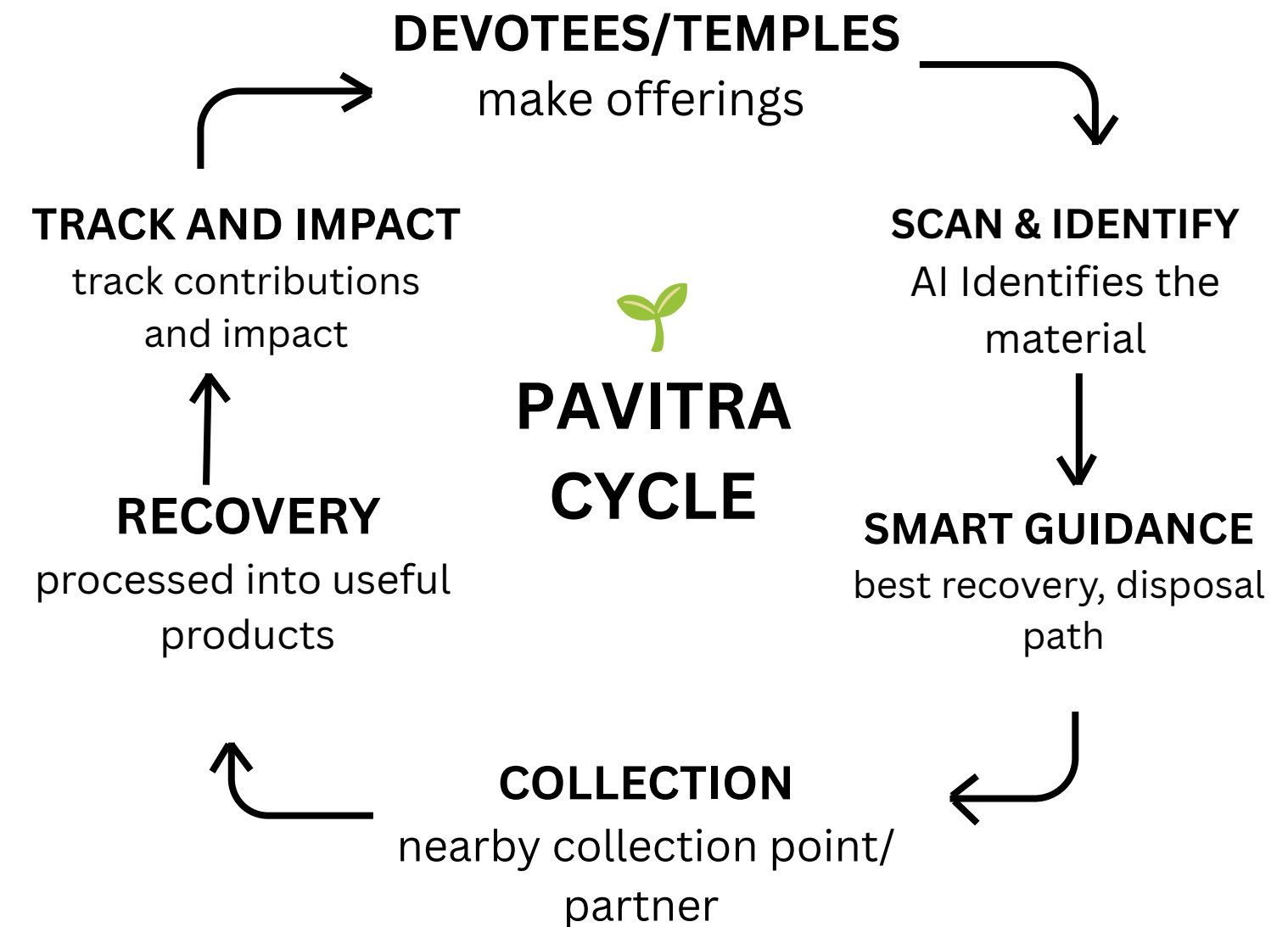
Potential Impact

- ▲ Prevents religious waste from reaching rivers and open areas.
- ▲ Connects people with verified disposal & recovery centres.
- ▲ Encourages responsible & respectful disposal.

Benifits

- ▲ **Environmental:** Cleaner water bodies and surroundings.
- ▲ **Social:** Preserves religious sentiments while promoting cleanliness.
- ▲ **Economic:** Supports recovery of materials into useful products.

Our Solution -Pavitra Cycle



1. CPCB- Revised Guidelines for Idol Immersion

<https://cpcb.gov.in/openpdf.php?>

[Bid=TGF0ZXN0RmlsZS8yOTVfMTU4OTI3Mzk4Nl9tZWRpYXBob3RvMjQzNjUucGRm](https://cpcb.gov.in/openpdf.php?Bid=TGF0ZXN0RmlsZS8yOTVfMTU4OTI3Mzk4Nl9tZWRpYXBob3RvMjQzNjUucGRm)

2. PIB, Ministry of Housing and Urban Affairs -

Flower Power: India's Temple Waste Transformation

<https://www.pib.gov.in/PressReleasePage.aspx?PRID=2056882®=48&lang=2>

3. PIB/ Ministry of Jal Shakti-

Pollution in Rivers and Religious Waste

<https://www.pib.gov.in/Pressreleaseshare.aspx?PRID=1795095&lang=2®=3>